Supplemental Material:
Strong Optical Disorder Erases Images but Preserves Local Change Observability

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This Supplemental Material is an audit trail for the Letter. It contains (S1) the general Chernoff/Kullback–Leibler jet derivation and its equality cases; (S2) the JET_TEST design with all seventeen frozen checks and the raw exponent tables; (S3) the full-scramble Gaussian derivation, exact generator, rank-one result, and detection duel; (S4) the sealed thin-screen protocol with bars D0–D6, the fixed/fresh comparator, the D3/D5 boundary failures, and the chunking audit; and (S5) the thin-screen covariance aperture and a minimal prior-art fence. All numerical values are read from the committed frozen artifacts `JET_TEST` (`1bf29f1`), `SCRAMBLE_EXT` (`ed7a1e0`), and `SEALED_DET/CONFIRMATORY` (`b37c841`); no new data are generated here.

GENERAL QUOTIENT-JET DERIVATION AND EQUALITY CASES

Let $P_{\varepsilon,\eta}$ be a dominated statistical model in which ε is a scalar physical-change amplitude along a scene direction δ and η is a nuisance. Define the profiled Chernoff distance

$$C_*(\varepsilon) = \inf_{\eta'} \max_{0\le s\le 1} \Big[-\log \int p^{s}_{\varepsilon,\eta}\, p^{1-s}_{0,\eta'} \Big], \tag{S1}$$

and assume the infimum is attained locally with

$$C_*(\varepsilon) = c_m\, \varepsilon^{2m} + o(\varepsilon^{2m}), \qquad c_m > 0. \tag{S2}$$

Theorem A (information-jet detection law). For T independent observations with equal priors,

$$\lim_{T\to\infty} -\tfrac{1}{T} \log P_e^*(T,\varepsilon) = C_*(\varepsilon), \tag{S3}$$

so in the joint small-change/large-sample regime with target error exponent h,

$$T_{\rm req}(\varepsilon) = \frac{h}{c_m\, \varepsilon^{2m}}\, [1 + o(1)]. \tag{S4}$$

For quickest detection at false-alarm level α, if the profiled per-bank KL distance is $D_*(\varepsilon) = k_m \varepsilon^{2m} + o(\varepsilon^{2m})$, then Lorden/CUSUM asymptotics give the mean delay $E_{\rm delay} = |\log\alpha|/(k_m\varepsilon^{2m})[1 + o(1)]$. The exponent m is invariant because reparameterizing η leaves the nuisance orbit unchanged, a gain gauge absorbed into η is quotiented automatically, no statistic or Fisher eigenbasis is selected, and physical covariation changes c_m but not the definition of m.

Gaussian specialization. Let $P_\varepsilon = \mathcal{N}(0, \Sigma_\varepsilon)$ with $\Sigma_\varepsilon = \Sigma_0 + z(\varepsilon)H$, $\Sigma_0 \succ 0$, and put $A_\varepsilon = z(\varepsilon)K$, $K = \Sigma_0^{-1/2} H \Sigma_0^{-1/2}$. The exact divergences are

$$D(P_\varepsilon \| P_0) = \tfrac{1}{2}\big[\operatorname{tr} A_\varepsilon - \log\det(I + A_\varepsilon) \big] = \frac{z^2}{4}\operatorname{tr}(K^2) - \frac{z^3}{6}\operatorname{tr}(K^3) + O(z^4), \tag{S5}$$

$$B(P_\varepsilon, P_0) = \tfrac{1}{2}\log\det\Big(I + \tfrac{A_\varepsilon}{2}\Big) - \tfrac{1}{4}\log\det(I + A_\varepsilon) = \frac{z^2}{16}\operatorname{tr}(K^2) + O(z^3). \tag{S6}$$

For nearby regular distributions the optimizing Chernoff parameter tends to $s = \frac{1}{2}$, so Eq. (S6) is the leading Chernoff coefficient.

Equality and boundary cases. With $z(\varepsilon) = 2c\,\varepsilon + d\,\varepsilon^2$, $c = x^\top G\delta$, $d = \delta^\top G\delta > 0$: (i) fast class, $c \neq 0$: $D = c^2\|K\|_F^2\varepsilon^2 + o(\varepsilon^2)$, $m = 1$; (ii) curvature class, $c = 0$, $d > 0$: $D = \frac{1}{4}d^2\|K\|_F^2\varepsilon^4 + o(\varepsilon^4)$, $m = 2$; (iii) true local blindness only if $d = 0$ (a nullspace of G at coarse grain) or the covariance signature is entirely nuisance; (iv) global finite-change cancellation for signed changes at $\varepsilon_* = -2c/d$, so the theorem is local; on the monotone cone ($x \ge 0$, $\delta \ge 0$, G entrywise nonnegative) $c, d \ge 0$ and every $\varepsilon > 0$ gives $\Delta Q > 0$; (v) amplitude gauge: if $\Sigma = R + a\,Q(x)H$ then $\partial_a\Sigma = QH$ and $\partial_Q\Sigma = aH$ are collinear, so zero-lag covariance carries zero efficient information for separating Q from a. Under nuisance profiling with $\langle A, B\rangle_0 = \frac{1}{2}\operatorname{tr}(\Sigma_0^{-1}A\Sigma_0^{-1}B)$ and $H_\perp = (I - \Pi_\eta)H$, Eqs. (S5)–(S6) hold with $\|K_\perp\|_F$, $K_\perp = \Sigma_0^{-1/2} H_\perp \Sigma_0^{-1/2}$; if $H_\perp = 0$ the change is unidentifiable from that ledger.

JET_TEST DESIGN AND THE SEVENTEEN FROZEN CHECKS

The jet test uses the exact scrambling generator (Sec. S3) on a 32×32 grid ($N = 1024$), code band $P_B = 3$, grain band $K = 13$, $M = 36$ codes, and 10^4 photons per bucket, with $\|K\|_F^2 = 3.46 \times 10^{-4}$. Three scene directions are constructed: a generic beyond-band direction δ_g with $c = x^\top G\delta_g \neq 0$; an exactly G-orthogonal direction δ_o with $x^\top G\delta_o = 0$; and a small-eigenvalue direction δ_n. A mixed direction δ_{mix} with a small nonzero c probes the crossover.

Table S1 lists the raw contact orders and coefficient ratios. The exact-KL log–log slopes are 2.038 (δ_g) and 4.000 (δ_o, δ_n), matching the predicted integer orders; the measured leading coefficients agree with Eqs. (S5)–(S6) to 1.07% (δ_g) and 0.00% (δ_o). The Monte Carlo single-look $d' \propto \varepsilon^m$ exponents are 0.95 and 2.05; the CUSUM ($\text{ARL}_0 = 500$) delay slopes are -2.16 and -3.92. The mixed direction crosses over at $\varepsilon_{\text{cross}}^{\text{pred}} = 9.995 \times 10^{-3}$ versus empirical 9.997×10^{-3} (ratio 1.0003).

TABLE S1. Raw jet exponents (committed `JET_TEST`, `1bf29f1`). $c = x^\top G\delta$, $d = \delta^\top G\delta$; KL slope is the log–log regression of the exact Gaussian KL over an amplitude decade; coef. ratio is measured/predicted leading coefficient.

Direction	c	d	m_{pred}	KL slope	Bhatt. slope	coef. ratio
δ_g (generic)	29.78	164.26	1	2.0381	2.0374	1.0107
δ_o (orthogonal)	2.0×10^{-15}	137.43	2	3.9999	3.9999	1.0000
δ_n (small eig.)	-3.1×10^{-6}	44.19	2	4.0000	4.0000	0.9999

TABLE S2. Monte Carlo and sequential-detection slopes, and the nuisance quotient (committed `JET_TEST`). I_θ is the per-bank efficient information for log fluctuation scale; "naive" omits nuisance profiling.

Quantity	δ_g / value	δ_o / value
MC d' exponent m	0.955	2.046
CUSUM delay slope	-2.158	-3.916
Amplitude gauge, single zero lag: I_θ^{naive}	0.614	
Amplitude gauge, single zero lag: I_θ^{eff}	0.000	
Amplitude gauge, proportional lags: I_θ^{eff}	0.000	
Second grain kernel: fraction recovered	0.018	
Amplitude anchor: fraction recovered	0.200	

The signed iso-Q cancellation crosses at $\varepsilon_* = 0.363$ with $\text{AUC} = 0.481$ (chance), while the two sides remain visible ($\text{AUC} = 0.026$ and 1.000), confirming locality. The monotone cone gives $\Delta Q > 0$ for all seven tested amplitudes (ΔQ from 1.59 at $\varepsilon = 0.005$ to 156.0 at $\varepsilon = 0.4$), with the kernel entrywise nonnegative. A scan of 400 in-aperture directions gives $d_{\min} = \delta^\top G\delta = 0.254$ (median 0.265), so the blind fraction is exactly 0. All seventeen frozen checks pass and none of the four kill conditions fire (Table S3; verdict `MOONSHOT_SURVIVES`).

FULL-SCRAMBLE GAUSSIAN DERIVATION, GENERATOR, RANK-ONE RESULT, AND DUEL

A band-limited modulator imprints a real, nonnegative code p_i (Fourier support B_p) on the illumination; a dynamic strongly scattering medium with state T_t produces the speckle field $E_{i,t}(r) = \sum_a T_t(r,a)A_i(a)$, and a single-pixel bucket collects $b_{i,t} = \sum_r |E_{i,t}(r)|^2 x(r) + \text{shot}$. Fully developed speckle means T_t has zero-mean circular-complex-Gaussian entries, so $E_{i,t}$ is a zero-mean circular-complex-Gaussian field with mutual coherence $J_{ij}(r,r') = \langle E_{i,t}(r)E_{j,t}(r')^*\rangle$. With medium statistics $\langle T_t(r,a)T_t(r',b)^*\rangle = C(r,r')\,\Gamma(a,b)$ and full scrambling $\Gamma(a,b) = \tau(a)\delta_{ab}$, the coherence factorizes,

$$J_{ij}(r,r') = C(r-r')\,O_{ij}, \qquad O_{ij} = I_0 \sum_a \tau(a)p_i(a)p_j(a). \tag{S7}$$

The joint field is exactly generated by $E_t = B(Z_t O^{1/2})$ with Z_t i.i.d. unit circular-Gaussian, $O^{1/2}$ the code-covariance root, and $BB^\dagger = C$; this Kronecker generator drives the simulation with no approximation.

TABLE S3. All seventeen predeclared JET_TEST checks (committed `JET_TEST`, `1bf29f1`); every entry is `true`, and each of the four kill conditions is `false`.

Check	Status
1. KL contact order generic = 2	pass
2. KL contact order orthogonal = 4	pass
3. orders integer	pass
4. generic coefficient within 10%	pass
5. orthogonal coefficient within 10%	pass
6. crossover within 25%	pass
7. delay slope generic near -2	pass
8. delay slope orthogonal near -4	pass
9. MC generic m near 1	pass
10. MC orthogonal m near 2	pass
11. amplitude gauge collapse to zero	pass
12. collapse persists across lags	pass
13. information returns with anchor	pass
14. AUC returns to $\frac{1}{2}$ at ε_*	pass
15. locally visible either side of ε_*	pass
16. cone $\Delta Q > 0$ for all tested	pass
17. no broad in-aperture blind set	pass
Kills (non-integer order; ε^{-4} class vanished; info survives collinear amplitude; broad blind set)	all `false`

Mean wall. The mean intensity is $\mu_i(r) = J_{ii}(r,r) = C(0)O_{ii}$, independent of r, so $\mathbb{E}[b_i] = C(0)O_{ii}\,\hat{x}(0)$: the first-moment channel measures only the scene DC term, with spatial band $B_{\text{mean}} = \{0\}$. Every h with $\hat{h}(0) = 0$ is annihilated. Numerically the mean envelope coefficient of variation is 0.0156 (Monte Carlo floor 0.0158), and the mean response to a beyond-band, DC-free change has relative magnitude 6.56×10^{-19}.

Covariance and rank collapse. By the Siegert relation $\mathrm{Cov}_t[S_{i,t}(r), S_{j,t}(r')] = |J_{ij}(r,r')|^2$, so with Eq. (S7) the same-epoch bucket covariance collapses to

$$\mathrm{Cov}_t[b_{i,t}, b_{j,t}] = O_{ij}^2\, Q(x), \qquad Q(x) = \sum_{r,r'} x(r)x(r')|C(r-r')|^2 = x^\top G x. \tag{S8}$$

In Fourier, $Q(x) = \sum_k \hat{G}(k)|\hat{x}(k)|^2$ with $\hat{G} = \widehat{|C|^2}$ supported on $|k| \le 2k_{\text{grain}}$. The measured aperture is full ($\hat{G}$ nonzero on all 1024/1024 bins, $C_0 = 0.517$), yet the scene enters only through the scalar Q, so the covariance Fisher information is rank one, $I^{\text{cov}}_{xx} = 4\kappa\,(Gx)(Gx)^\top$. Empirically $\hat{Q}$ from $\mathrm{Var}[b_i]/O_{ii}^2$ is 190.9 against the theoretical $Q(x) = 187.4$ (ratio 1.019), and 99.99% of the covariance matrix energy lies in the single direction $O\circ O$. The change functional obeys $\Delta Q = 2x^\top G\delta + \delta^\top G\delta$; the predicted $\Delta Q = 0.34803$ matches the measured 0.34803 to relative error 2×10^{-15}, split into cross term 0.0031 and quadratic term $0.3449 > 0$ for the orthogonal direction.

Detection duel. Table S4 reports the H0-vs-H1 duel (efficient covariance score) at the complete-scramble endpoint. The mean detector is at chance for both a coherent and an orthogonal beyond-band 5% change (AUC = 0.502 and 0.505), while the covariance detector reaches $d' = 4.10$, AUC = 0.997 for the generic 5% change at 8192 banks. The worst-case orthogonal direction obeys the predicted $d' \sim \varepsilon^2\sqrt{T}$ law and is never zero; its slower rate ($a = 0.0050$ per $\sqrt{T}$ vs 0.0459 for the coherent change) is the curvature-class ε^{-4} price, not a blind direction.

SEALED THIN-SCREEN PROTOCOL, BARS D0–D6, COMPARATOR, AND AUDITS

The sealed sentinel uses a true zero-dimensional bucket record on a 64×64 scene ($N = 4096$), modulator band $k_p = 5$, $M = 128$ signed effective codes realized by 256 complementary nonnegative exposures per bank at a 20 kHz pattern rate (12.8 ms per bank; cap $T_{\text{eff}} = 4096$ banks = 52.4 s), 10^4 detected photoelectrons per physical bucket, and a medium quasi-static within a bank and independent between banks. Thresholds were committed before confirmatory

TABLE S4. Complete-scramble detection (committed `SCRAMBLE_EXT`, `ed7a1e0`). d' and AUC at listed bank budgets $T_{\rm eff}$; the mean detector uses 512 banks. “coherent” is a generic direction ($x^\top G\delta \neq 0$); “orthogonal” is G-orthogonal.

Channel / change	$T_{\rm eff}$	d'	AUC
Mean, coherent 5%	512	0.030	0.502
Mean, orthogonal 5%	512	0.015	0.505
Covariance, coherent 5%	512	0.99	0.762
Covariance, coherent 5%	2048	2.22	0.943
Covariance, coherent 5%	8192	4.10	0.997
Covariance, coherent 2%	8192	1.60	0.887
Covariance, orthogonal 5%	8192	0.36	0.601
Covariance, orthogonal 5%	65536	1.23	0.809

generation; the run used 12 confirmatory banks and 2.76 of a 6.0 GPU-hour ceiling. Table S5 summarizes the seven bars.

TABLE S5. Sealed detector bars D0–D6 (committed `SEALED_DET/CONFIRMATORY`, `b37c841`; analysis freeze `5910277`). “passed” is the frozen bar verdict; D3 and D5 did not pass their strong forms but did not trigger a kill, defining the honest boundary (Sec. S4).

Bar	Key measured endpoint	Verdict
D0 mechanism/engine	mean deriv. 5.7×10^{-16}; two-engine 1.7%; shot ledger 1.0%	pass
D1 calibration	$n = 108$ cells, median $d'_{\rm emp}/d'_{\rm an} = 0.998$, all in $[0.91, 1.10]$	pass
D2 capability	2% in 77/81 cells, best 453 banks, MC power LCB 0.990	pass
D3 specificity	TPR 0.988, bal. acc. 0.916; medium FA $0.096/0.084 > 0.05$	boundary
D4 fresh comparator	fixed vs fresh latency 459 vs 1013 banks (retain fixed)	pass
D5 mismatch	AUC 1.000; $\leq 10\%$ rotation passes; 20%/slope-1 fail	boundary
D6 online feasibility	0.0203 ms (0.16% of bank), 0.26 MB	pass

D2 capability. The 2% beyond-band change is analytically detectable within the 4096-bank cap in 77/81 predeclared cells; the best cell (unit static factor, k^{-2} spectrum, $k_w/k_p = 1$, $1.25k_p$ claim shell) requires 453 banks and the pooled Monte Carlo power lower bound is 0.990 ($n = 13500$). The 5% change clears all 81 energy-spread cells and 75/81 worst-mode cells; the 1% best cell clears $d' \geq 5$ by 1811 banks; the 0.5% edge audit reaches $d'_{\max} = 3.76$.

D3 specificity (strong claim killed, power intact). At the threshold calibrated to 1% H0 false alarm ($t = 0.051$), the target true-positive rate is 0.988, in-band false alarms are 0.020, and four-class balanced accuracy is 0.916; the beyond score stays centered on non-targets ($|d'| \leq 0.019$) and the attribution-simplex maximum off-diagonal canonical correlation is 0.039. However, medium-amplitude and medium-timescale events exceed the frozen 0.05 false-alarm bar at 0.096 and 0.084, and the weakest class-specific separation is 1.79 rather than 5 (the intended per-class separations are [1.79, 43.06, 4.78]). The result is a power-certified beyond-band alarm with measured attribution leakage, not a fully specific sentinel; no “specific”, “zero-false-alarm”, or “medium-immune” wording is used.

D4 fixed-versus-fresh. On identical photons, exposures, duration, anomaly bank, and law knowledge, the repeated (fixed) code bank retains a latency advantage: best cell 459 (fixed) versus 1013 (fresh) banks, mid cell 1271 versus 1488, floor cell 1598 versus 3334; the frozen branch retains the fixed-code concentration.

D5 mismatch (robustness frozen to $\leq 10\%$ rotation). AUC is 1.000 on every tested axis and latency inflation is bounded, but false-alarm calibration fails outside the primary domain: at aperture-preserving basis rotation 10% the non-target false alarm is 0.032 (latency inflation 19%, pass), whereas at 20% rotation it is 0.052 and at spectral-slope mismatch -1 it is 0.076 (fail). Robustness is therefore frozen to the $\leq 10\%$ rotation regime; the kill-tree verdict is “narrow the physical domain or kill robustness wording”, and the broader wording is withdrawn.

Integrity audits (verbatim). Record generator. After protocol freeze and before unblinding, the record generator was refactored solely to stream Monte Carlo records in GPU-memory chunks; an audited diff found no changes to hypotheses, thresholds, bank counts, scenes, estimators, or endpoints, although altered random-number consumption produced a different, still-blinded realization set. All reported decisions apply the frozen analysis unchanged to that disclosed chunked stream. *Shot model.* A mandated independent-engine check exposed a signed-code shot-noise error that had inflated covariance Fisher information by approximately 1.9×; replacing the signed difference Px with the physical complementary-exposure throughput $|P|x$ reduced the frozen $P1$ witness coverage from 0.050 to 0.033 and eliminated the apparent natural-scene living pocket, so all values reported here come from the corrected artifacts and every prior kill only strengthened.

THIN-SCREEN COVARIANCE APERTURE AND MINIMAL PRIOR-ART FENCE

Aperture interpolation. The thin multiplicative screen is the weakest complex medium: a single random phase layer w_t. Its bucket covariance is $C(x) = \Phi^2 P \operatorname{diag}(x) K_w \operatorname{diag}(x) P^\top$, whose informative support is the convolution aperture $B_w \oplus B_p$ and whose Fisher structure is rich—many recoverable scene modes. As the medium approaches complete scrambling, the input coupling Γ loses angular memory (rank $r_{\mathrm{ME}} \to 1$) and the covariance factorizes as in Eq. (S7), collapsing all $M(M+1)/2$ entries to the single scalar $Q(x)$. For a residual memory effect of rank r_{ME} the covariance becomes a rank-$O(r_{\mathrm{ME}}^2)$ combination of quadratic scene functionals, so the recoverable imaging aperture interpolates monotonically from the rich thin-screen aperture to the rank-one scalar. *Beyond-band imaging survives exactly to the extent the medium retains memory-effect modes; beyond-band detection survives complete scrambling*, because $\delta^\top G \delta > 0$ for every nonzero δ. This is the precise, physically correct boundary between the two disorder endpoints of the Letter.

Minimal prior-art fence. The result is distinct from the head-on prior art. Memory-effect and speckle-illumination imaging through scattering—blind structured illumination and its superresolution analyses [1, 2] and photoacoustic speckle-fluctuation imaging [3]—*reconstruct* an image by exploiting residual correlations; our contribution is the opposite regime, where reconstruction rank has collapsed to one and only the observability order survives. Quantum-inspired superresolution [4] moves Fisher information between optical *measurements* of the same scene; the Rank–Jet Separation moves contact order between *statistical layers* of one scalar record, with no change of hardware. Speckle-decorrelation and coda-wave change detection [5] sense a change in the *medium*; here an exact first-moment invariance wall makes the medium a transducer for *scene* change while a profiled quotient rejects medium-law events. No claim of prior-art equivalence is made, and the novelty is the nuisance-profiled statistical jet order as a physical observability class for disordered sensing.

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